Ex 12:

implement a Deep convolutional GAN to generate complex color images

Aim:

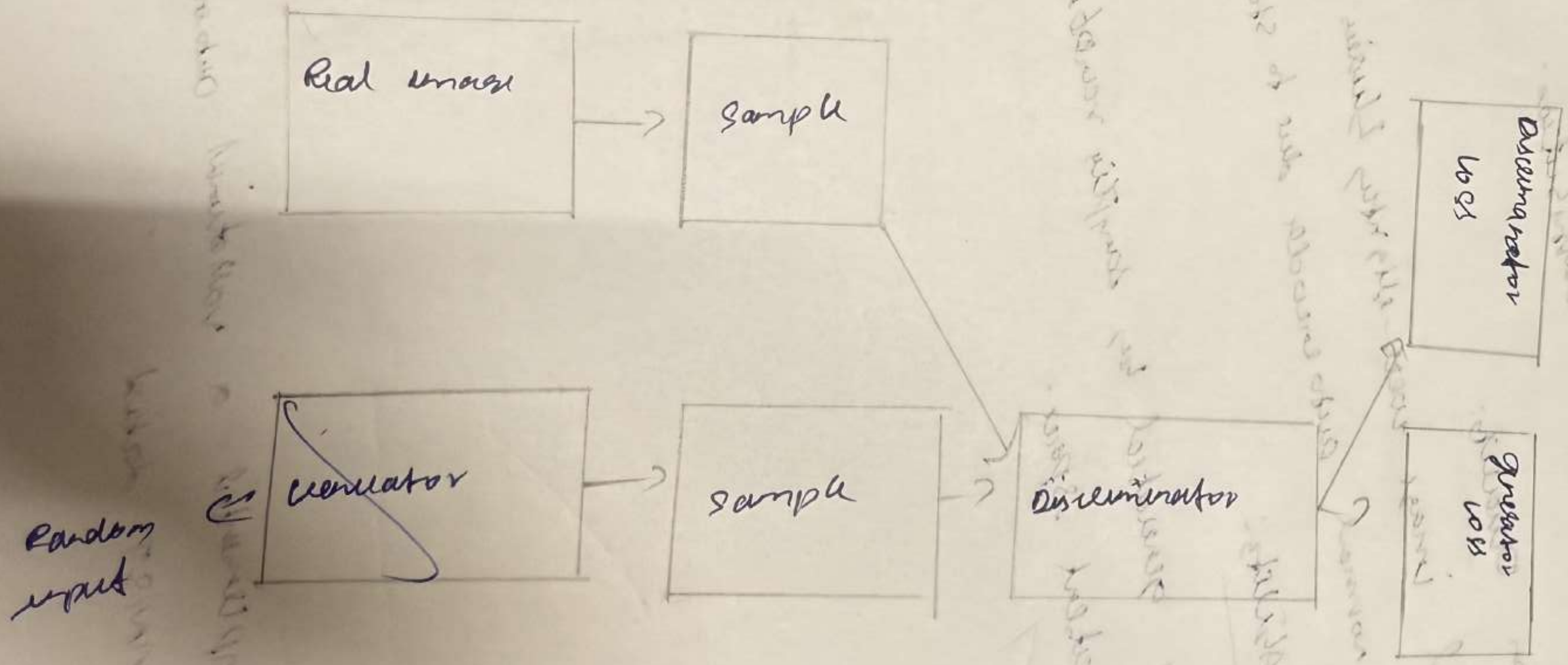
To implement and train a Deep convolutional GAN (Generative Adversial network).

Objective:

* To understand GAN Architecture
* To train model on color image dataset
* To observe training dynamics and Quality of generating images.

Pseudocode:

1, import libraries and set device
2, Load CIFAR-10 dataset, normalize to [-1, 1] create dataloader.
3, Define DCGAN Generator: series of convtranspose 2D, Batch norm 2d, ReLU.
4, (mean = 0, std = 0.02)
5, Define Loss & optimizers.
6, for each epochs -
   a, train Discriminator on real images
   b, save generator & batch of generate image


Real image
sample
Generator
sample
Random input
Discriminator
Discriminator loss
generator loss

Observation:

+ The discriminator and generator losse usually fluctuate This is expected in GAN training as one improves, other responds.

+ initially generator images are noisy random patterns.

+ later epochs starts showing structure colours, shape).

+ Saving image each epoch helps visualize from noise -> structure -> clear image

Result:

successfully implemented DCGAN

outputs

Epoch (1/5) Loss D: 0.9934 Loss G: 2.9012

Epoch (2/5) Loss D: 0.5894 Loss G: 2.66182

Epoch (3/5) Loss D: 0.7406 Loss G: 2.3580

Epoch (4/5) Loss D: 0.5540 Loss G: 2.0222

Epoch (5/5) Loss D: 1.3862 Loss G: 1.5855